

SCHEDULE - I "

TECHNICAL SPECIFICATIONS FOR DESIGN, MANUFACTURE, TESTING AT SHOP AND SUPPLY, INSTALLATION & COMMISSIONING OF UPS SYSTEM, ITS ACCESSORIES AND ALLIED WORKS AT SITE:

The UPS being the own design of the UPS vendor, it shall be suitably designed manufactured, tested at shop and supplied. The design standards, materials or components used manufacturing process shall all conform to the respective BIS Code and any other standards mentioned elsewhere in this tender as applicable. However, the following specifications, technical parameters, etc, are furnished by way of basic or minimum requirement for the end result. The tenderer shall be solely responsible for the entire design/ specification, safety aspects of the entire UPS system and related works as detailed in the tender notwithstanding the specifications furnished hereunder.

A. SCOPE OF SUPPLY:

1. **The Main System:** ON-LINE Un-interrupted Power Supply (UPS) System comprising of 1 set of 5 + 5 KVA UPS with hot standby configuration as per the details indicated in the price bid, battery source, related software for interface compatible with lan / unix system shall be designed by the manufacturers own technology and standards but it shall also meet the required parameters detailed below apart from the standard design parameters of the manufacturers. They shall be manufactured, tested at factory for its quality and performance. Such units shall be supplied, installed and commissioned at site and undertaking allied works like interlinking UPS system with battery, input power supply to the UPS system, output etc.

2. Hot standby Operation:

In nutshell, the whole system comprising of one set 5 + 5 KVA UPS systems should be configured in hot standby mode with common battery bank with iron stand. **WITH Supply, Install, Commission & Testing UPS COMPATIBLE WITH EXISTING (ALREADY PRESENT) 16 Nos 100 Ah SMF Batteries .**

For Hot standby operation of one set 5 + 5 KVA UPS systems, where load is taken up by first UPS, in case of failure of first UPS the load should be fully taken over the other, without interruption.

If both the UPS fails, the loads should be transferred to bypass without interruption in power supply.

The design of the UPS should be such that where shall not be any common parts in the circuit, which will lead to single point failure.

It should be possible to isolate the failed UPS for repairs without the power output of other UPS.

3. System Maintenance:

Each UPS system will be serviceable independently and at the same time at-least one UPS system should be available to serve load. **Any scheduled maintenance actively will be carried out without disrupting load at any point of time.**

4. ACCESSORIES:

Copper cable of 90% and above purity of appropriate size and current carrying capacity between battery source and UPS system should be used. The cable size, the current carrying capacity should be highlighted. The type of copper cable should be mentioned. Any other unspecified but related to installation of UPS system, materials required for fixing the UPS system.

5. INSTALLATION & COMMISSIONING:

Moving, unloading & Positioning of UPS system alongwith necessary electrical, batteries and interconnections etc. Cabling between batteries to UPS system and between UPS system to the input power source and output power point. Necessary pre-commissioning, testing and load testings for 24 hours upon installation should be conducted by the successful vendor.

6. Drawings & Manuals : Being a self designed electrical equipment, preparation and submission of drawings, submission of users manual, operation and preventive measures to be taken etc., shall be furnished in duplicate and all other details that are essential for its operations.

7. Site Conditions: The various aspects and the condition of the site and layout needs to be well understood in executing the specified work. The input power to the UPS system could be either by way of Electricity Board Power or by means of DG Power. The UPS system shall be capable of receiving either source of power at fluctuating voltage and frequency and deliver the design output conditions as specified in para B of this schedule. The location will be within the specified City limits and with its inherent conditions.

8. Bank's Scope of Work : The Bank shall arrange for input power supply with necessary input supply. Necessary input and output cabling up to the designated point of UPS location, and output distribution cabling and control system will be undertaken by the bank. However the necessary assistance & guidance shall be under the scope of the supplier. Existing earthing arrangement shall be extended to the system.

9. The below mentioned specifications furnished are broad guidelines for the UPS system base` on the tenderer's own design and manufactures standard system. Wherever the specifications are not furnished or specific, the specification shall be as per their respective IS standards / ISO standards or as per best engineering practice and they shall not be diluted.

**(B) TECHNICAL SPECIFICATIONS FOR 5 + 5 KVA DIGITAL SIGNAL PROCESSING(DSP)
CONTROLLED ON LINE IGBT UPS SYSTEM:**

Specifications	Bank's Requirement
Technology	a. UPS systems with DSP controlled IGBT based rectifier technology in True on line configuration with double conversion using IGBTs in the inverter and converter. b. Provision for configuring 2 UPS systems in parallel load sharing mode. c. UPS should consist of converter/ Rectifier, Battery Charger, Static Inverter with static switches, Static bypass switch and isolation transformer d. Input Active power factor correction
Rating (output power max.)	5 + 5 KVA UPS System with common battery bank
Power Consumption	Full Load - 8.0 KW & 35 Amps 75% Load - 7.5 KW & 32 Amps
Model Name Model No. Manufactured by	
INPUT Phase Nominal Input Voltage Nominal Input Voltage Range Frequency Nominal frequency Range	Single Phase 230 Volts AC + or - 25% AC (on full load) 50 HZ 45 Hz to 65 HZ

Input Power Factor with auto input P.F correction system DG Compatibility Input over voltage protection Input under voltage protection Rectifier Input Current THD	Better than 0.96 Should be provided Should be provided Should be provided IGBT Less than or equal to 4%
OUTPUT Phase Output Voltage Output Voltage Range Frequency DG Compatibility Rectifier Input Current THD	Single Phase 230 Volts AC + or - 1 % V AC (on full load) 50 HZ Should be provided IGBT Less than or equal to 5%
Isolation Transformer	UPS output is isolated from mains input and battery (Galvanic Isolation Transformer)
Audible Alarm for a) Battery Low b) Mains Failure c) UPS Failures Battery re-charge time from fully discharged to 100% charged condition Cold start	Required Required Required Should not exceed 12 hours The facility to start the UPS even if main power is not there at the time of starting. The system should draw current from the batteries and start. This should be demonstrated at the time of installation.
Charger Type Voltage Regulation Ripple(Without Batteries) Battery Recharge Time from fully discharged condition to 100% charged condition.	Constant Voltage, Constant Current (The charger should be designed for atleast 15% of total battery current). <u>The charger should be preferably temperature compensated with automatic battery test.</u> +/- 1% < 1% Should not exceed 12 hours
Transient Response Response Time	+/- 5% (100 % load step) Less than 60 mili seconds
Automatic Static by-pass switch facility to mains in case of UPS failure	Should be provided & should take care of 100% uninterrupted transfer of load from UPS (synchronous bi-directional static switch with zero transfer time).
Manual By-Pass Switch	Built-in Manual By-Pass should be provided
Overall Efficiency(AC to AC)	Greater than 92 % in online mode
Display Panel	Built in LCD/LED Display Input Voltage Output Voltage

	Heat Sink Temperature Battery Voltage Output Current (%) Output Frequency etc
Indications	AC Mains line Battery on Mains abnormal/ out of range Load on UPS Load on battery Inverter mode Load on Auxiliary/ bypass Inverter trip
Protection for (Audible alarms should be provided for important protections)	Input MCB Input single phasing/ reversing Rectifier Over/ under Voltage Battery MCB Battery Low Battery Charging Current Limit Output under Voltage Output Over voltage, Inverter over Temperature Output Overload Output Short circuit etc, overheat Surge Protection
Interface facility	The UPS system should have necessary hardware and software a) RS 232 to work on existing operating systems b) Remote Management. There should be a facility to monitor and broadcast to all workstations, whenever necessary, conditions such as 1. Power failure: UPS working on batteries preferably to indicate present load in percentage, time up to which batteries can support with present load & warning well in advance of shutdown of the system. 2. The Software should be capable of automatically closing files so that data on our computers is not lost/corrupted.
Environmental Operating Temperature Storage Temperature Humidity Audible Noise Level Cooling Enclosure Protection	0 - 45 Deg C (Continuous) 0 - 55 Deg C 10-90% (Non Condensing) < 52 dB at full load upto distance of 1 mtr forced air IP 20
Warranty	UPS should have 2 year warranty from the date of supply
External Testing Certificate	Must

SPECIAL CONDITIONS AND DETAILS OF TESTING TO BE UNDERTAKEN FOR THE WORK OF SUPPLY, INSTALLATION, & COMMISSIONING OF 5 + 5 KVA UPS SYSTEM :

1.0. Time is The Essence of Contract : The tenderer are requested to note that the UPS system is to be provided for Bank's sensitive equipments and if for any reasons the work of supply & commissioning is delayed the Bank's reputation and commitments will be affected and penalty stipulated as per liquidated damages will be levied with-out any requirement on the part of the Bank to establish the actual loss in monetary terms. This aspect shall be the terms of this tender and will be binding on the tenderers.

2.0. Power Consumption: Being a techno-economic project, the power consumption, overall efficiency of the UPS system, AMC charges and annual recurring expenditure shall be assessed and these aspects shall be one of the criteria in evaluating the cost economics besides the initial capital cost. Therefore the tenderer shall furnish the required particulars to establish the economics. The rated performance shall be subject to testing and any shortfall shall be liable for penalization.

The details of power consumption of the entire UPS system at input level, including the battery charging power shall be furnished for at full load conditions **AND** at 75 % load conditions Further, its corresponding Invertor efficiency and overall efficiency shall also be furnished.

3.0. The tenders shall quote for a **TWO YEAR WARRANTY** for all the equipment quoted by them.

4.0. MAINTAINANCE OF UPS EQUIPMENT

A) MAINTAINCE DURING DEFECT LIABILITY PERIOD OR WARRANTY PERIOD

The tenderer shall maintain the UPS Equipment free of cost and ensure that it works as per tender parameters. All spares required for normal operation as per tendered parameters shall be replaced at no extra cost. The number of visits shall be same as stipulated in Comprehensive Annual Maintenance.

B) COMPREHENSIVE ANNUAL MAINTAINANCE. (CAM) (after the defect liability period)

The tenderer shall maintain the UPS equipment for a period of **Five years** after the expiry of defect liability period and ensure that it works as per the tendered parameters. The rate of annual maintenance charges shall be furnished for a period of 5 years after the expiry of warranty/defect liability period and the rates quoted shall be binding on the tenderer.

5.0 The testing & performance parameters :

The following pre-delivery test shall be conducted at factory /manufacturing place and necessary test certificates shall be furnished before dispatch for all the technical parameters including the full load test as specified.

- 1>. Line Regulation Test & No- load regulation test
- 2>. Load Regulation Test at full, & 75% load.
(on both resistive & reactive load @ 0.8 pf)
- 3>. Overloading Conditions
- 4>. Invertor Efficiency at full load.
- 5>. Overall Efficiency at full load.
- 6>. Transient Response
- 7>. Response (Recovery) Time
- 8>. Testing of Protection; Short Circuit, Under / Over Voltage conditions.
- 9>. Total harmonic distortions.
- 10>. Load Crest Factor.
- 11> Wave form test on resistive and inductive load

12> Noise level test

13> Battery backup time cutoff and alarms test at site.

After, physical installation, the UPS shall be tested and commissioned at site by the manufacturers engineers such as voltage, frequency, input current, harmonic distortion, output voltage harmonic distortion, input power factor, efficiency, noise level, overload capacity as specified herein.

The power consumption shall be established at site by using **standard power analyzers** using external load brought by the tenderer for a continuous period of 24 hours. Load test shall also be established at site by using an external load to be brought at specified time as decided by the bank. All the activities under this clause shall be completed within the stipulated time schedule.

6. **EXTERNAL TESTING:** Testing of UPS should be carried out from reputed testing centers like Electronics Test and Development Centre, (ETDC) Dept. of Electronics, Govt. of India, or CPRI Bangalore or other reputed testing centre approved by the Government of India. The tenderer shall be liable to subject the UPS system to such testing.

The cost of such testing and transportation shall be borne by the UPS VENDORS ONLY. Also the cost of damages during transportation to the testing centre up & down, rectifications of the system or any of its components resulting out of such testings etc. shall be on the part of the tenderer and the employer shall not be responsible for any cost or loss incurred by the tenderer while undertaking the testings.

7.0. **Penalty Clause Against Deviation From Committed Parameters :** The tenderer are requested to furnish the technical parameters like overall efficiency, power consumption, capacity of the UPS system and other technical parameters after careful study. In case the actual parameters obtained at site and test results differs from those stipulated in the offers, the tenderer will have to be prepared for penalty for the shortfall in the capacity as well as higher power consumption. The following methodology shall be adopted for computing the quantum of penalty. A tolerance of 5% of the committed parameter in case of power consumption may be condoned by the accepting authority at his discretion.

In case of other parameters, as the UPS system is critical in nature, if the shortfall is likely to result in un-reliable operating conditions on " ON LINE " concept, the entire UPS may be rejected and the contractor is liable for repayment of entire amount paid by the employer and the Initial Security Deposit is liable for forfeiture. This is without any prejudice to initiate any legal action by the Bank and damages eligible for. The Bank has the option to revoke the Bank Guarantee.

7.1. **Capacity of the Equipment:** In case of any shortcomings, the difference between the committed capacity and the actual capacity measured at site or external testing centre shall be guiding factor to determine the quantum of penalty. The pro-rata cost reduction shall be effected by the following formula:

$$\text{Quantum of penalty} = \frac{\text{Cost of UPS system (Note-1)} \times \text{Difference in capacity} \times \text{Penalty Factor (*N2)}}{\text{Committed Capacity of UPS system}}$$

Note : 1). Cost of UPS system excludes battery cost & software cost.

2). Penalty factor will be 1.5 times.

7.2. **Power Consumption:** In case of short comings in power consumption, the difference between committed value and actual value will be converted into Units per hour (KWH). Excess power consumed over a period of 6 years at 8 hours per day on full load condition for 300 days an year shall be assessed and 20% of such assessed quantum will be liable for recovery. The actual consumption of power will be subject to assessment made by the consultant / Bank Engineer in the similar manner while undertaking the assessment made while evaluating the project. The accepting authority may at his discretion may evaluate in comparision with other offers and extend any or some relief which shall be binding.

Sl. no.	Item description	Qty & unit
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9.0 Departure From

Specification : Schedule of departure from the specification if any, shall be furnished by the tenderer in technical and commercial bid along with implication on the system and cost.

SCHEDULE III

SCOPE OF COMPREHENSIVE ANNUAL MAINTAINANCE CONTRACT

The tenderer shall offer the following at no extra cost to the bank. The rate quoted for CAMC should include the following.

1. The tenderer shall maintain the UPS System and ensure that it works as per the tendered parameters.
2. All the repairs and replacements of spares shall be carried out which are necessitated due to usage of system as per tender stipulations. However the repairs and replacements necessitated by loss or damage due to misuse or accident, fire or natural calamities shall be out of the scope of CAMC.
3. The tenderer shall inspect the equipment at least once in a month as a part of preventive Maintenance. This is apart from any breakdown visits that may be required which may emanate in between the two preventive maintenance inspections.
4. The maximum breakdown time shall be one hour from the time of receiving the complaint over phone. If the tenderer is not in a position to set right the defect or repair and put back the UPS to working condition within one hour, a standby unit of to suit the tender parameters should be provided immediately.
5. During the Preventive Maintenance the tenderer shall inspect/ check the equipment and record the following in log book, which shall be verified and confirmed by the bank's representative.
 - a. UPS System : Any malfunctioning of the components.
 - b. Any other remarks regarding the Electrical System:
 - c. Any steps to be taken by bank for working of UPS system.
6. During the preventive Maintenance inspection tenderer shall carryout the following apart from the other requirements for functioning of system as per tender parameters and same shall be recorded in a log book which shall be verified and confirmed by bank's representative.

UPS System : i. General cleaning. ii. Loose contact.
7. During the break down calls, the nature of repair carried out, parts replaced etc shall be recorded in the log book.

OTHER TERMS WITH REGARD TO CAMC:

The rate of annual maintenance charges shall be furnished for a period of 4 years after the warranty/Defect Liability Period and rates quoted shall be binding on the tenderer. The scope and terms of Annual Maintenance Contract like number of monthly visits, items covering under Annual Maintenance Contract, replacement of spares exclusions if any, down time for break down and service calls shall be as per as per the bank terms. The Bank reserves the right of discontinuing the Annual Maintenance Contract from the contractor during the tenancy of Annual Maintenance Contract. The cost of Annual Maintenance Charges shall also be one of the criteria in evaluation of cost economics of the tender.

The Annual Maintenance Charges shall be paid once in six months in advance subject to satisfactory maintenance for the period of previous six months and on production of bank guarantee equal to the amount paid in advance.

1.	Design, manufacture, testing at factory, supply, installation, commissioning and testing of state of the art 5 + 5 KVA capacity Single Phase input Single phase output Digital Signal Processing controlled, IGBT Rectifier Technology based online UPS system complying all technical specifications / features/ international standards as described in the tender document. With SNMPS interface facility.	1 set (5+ 5 KVA)
2	Supply, Install, Commission & Testing 16 Nos SMF 100 Ah Batteries	16 Nos of 100 Ah
3.	Comprehensive annual maintenance charges for the above set consisting of entire 5+ 5 KVA UPS AFTER DEFECT LIABILITY PERIOD OF 2 YEARS.	
	1 ST YEAR	
	2 ND YEAR	
	3 RD YEAR	
	4 TH YEAR	
	5 th YEAR	

The UPS vendor has to maintain the equipment supplied for at-least 10 years after the expiry of warranty period. At the same time, the vendor is expected to make available the spare parts for the systems for atleast 10 years after the expiry of warranty period. Thereafter, vendor will give 12 months notice prior to discontinuation of services. AMC amount for the five year after warranty period will be calculated to decide Lowest Vendor.

UPS SYSTEM WITH BATTERIES SHOULD BE SUPPLIED AT FOLLOWING PLACES:

S.NO	LOCATION	NEAR BY CITY
1	Churu (3824)	Churu
2	Hanumanagarh Town (5438)	Hanumanagarh Town
3	SadulShahar (5437)	
4	Suratgarh (5574)	Suratgarh
5	Karanpur (4234)	Churu
6	Sikar Ind Area (5575)	Sikar
7	Sriganganagar Main (2142)	Sriganganagar
8	Laxamangarh (4238)	Churu
9	Ladnu (3562)	Ladnu
10	Pilibanga (5439)	Pilibanga
11	Bikaner Kriti Samtabh (5298)	Bikaner
12	Srimadhopur (5440)	Srimadhopur